

Difference Between Type 1 and Type 2 Error

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Vikram Singh
Assistant Manager - Content

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Type-1 error is known as a false positive, i.e., when we reject the correct null hypothesis, whereas type-2 error is also known as a false negative, i.e., when we fail to reject the false null hypothesis. This article will discuss the difference between type 1 and type 2 errors.

Difference Between
Type-1 and Type-2 Error

	Null Hypothesis is TRUE	Null Hypothesis is FALSE
Reject null Hypothesis	Type I Error (False Positive)	Correct Outcome! (True Positive)
Fail to reject null Hypothesis	Correct Outcome! (True Negative)	Type II Error (False Negative)



Researcher/scientist assumes to prove or disprove their finding. These assumptions are also known as hypotheses. There are mainly two types of hypotheses **Null and Alternative Hypothesis**. Null and Alternative hypotheses are mutually exclusive statements. A null hypothesis statement is that there is no relation between the two variables. In contrast, an alternative hypothesis is a statement that refers to the statistical relationship between the two variables. While doing hypothesis testing, we encounter two types of errors, i.e., type-1 and type-2 errors. This article will discuss the difference between type-1 and type-2 errors. Type-1 and Type-2 errors are interconnected; reducing one can increase the probability of another. Type-1 error is a false-positive finding, while type-2 error is a false-negative finding in hypothesis testing.

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Difference between Type-1 and Type-2 Error

Parameter	Type -1 Error	Type – 2 Error
Definition	It refers to the non-acceptance of the hypothesis that ought to be accepted.	It refers to the acceptance of a hypothesis that ought to be rejected.
Represents	A false hit	A miss
Equivalent to	False Positive	False Negative
Notation	alpha	beta
Probability of committing an error	Equals to the level of significance	Equals to the power of test

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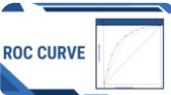
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Mean Squared Error



Mean squared error in machine learning

In this article we will talk about Mean squared error with example. It also explains how to calculate mean squared error. Mean squared error is calculated using python programming.



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Type -1 Error (Error of the first kind)

- It is also known as a false-positive.
- It occurs if the researcher rejects a correct null hypothesis in the population.
- Measured by alpha (significance level).
- If the significance level is fixed at 5%,
- **Cause of Type-1 Error**
- It can be reduced by decreasing the level of significance.



Difference between One-tailed and Two-Tailed Test

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Type -2 Error (Error of the second kind)

- It is also known as a false negative.
- It occurs if a researcher fails to reject a null hypothesis that is actually a false hypothesis.
- Measured by beta (the power of test).
- The probability of committing a type -2 error is calculated by $1 - \text{beta}$ (the power of test).
- **Cause of Type - 2 Error:**
- It can be reduced by increasing the level of significance.



Standard Error vs. Standard Deviation

Standard Error quantifies the variability between sample drawn from the same population, whereas the standard deviation quantifies the variability of values in a dataset. In this article we will discuss...[read more](#)



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Graphical representation of type – 1 and type – 2 Errors:



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Examples of Type-1 and Type-2 Errors

Now, let's take an example to understand better type – 1 and type – 2 errors:



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Example -1: A man goes to test the coronavirus (COVID-19). So, the possible errors are

Type -1 Error (False Positive): Test results are positive, but you don't.

Type – 2 Error (False Negative): Test results are negative, but you do.

Example – 2: A man goes to trial and is being tried for murder.

Null Hypothesis: Man is innocent until proven guilty.

Alternative Hypothesis: Man is guilty.

Type -1 Error (False Positive): Found guilty, but you are innocent.

True Positive: Found guilty and being guilty.

True Negative: Found Innocent and being innocent

Type -2 Error (False Negative): Found innocent, but you are guilty.

Key Differences

- When the null hypothesis is correct, and the researcher rejects it, this type of error is known as a type -1 error, whereas when the null hypothesis is false, and the researcher fails to reject it, this type of error is known as a type—2 error.
- Type -1 error is also known as false-positive, whereas type – 2 error is known as false-negative.
- We can reduce the type—1 or type—2 errors by decreasing or increasing the significance level.
- Type—1 and type—2 errors are inversely related; if one increases, the other decreases.

Difference Between Type 1 and Type 2 Error

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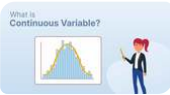
In this article, we have briefly discussed Type-1 and Type-2 errors and the difference between them using examples and graphical representation.

Hope, you will like the article.

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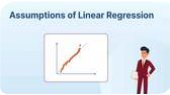
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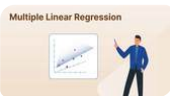
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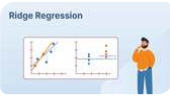
Assumptions of Linear Regression

There are certain assumptions of linear regression algorithm that must be satisfied before implementing over any model, otherwise it will lead to insignificant result.



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FAQs

What is the difference between type 1 and type 2 error?

Why is Type 1 error also called a false positive?

How is level of significance related to type 1 error?

What is the relationship between type-1 and type-2 error?

Can type-1 and type-2 error be avoided completely?

About the Author



Vikram Singh
Assistant Manager - Content

Vikram has a Postgraduate degree in Applied Mathematics, with a keen interest in Data Science and Machine Learning. He has experience of 2+ years in content creation in Mathematics, Statistics, Data Science, and Mac... [Read Full Bio](#)

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I have the following example: How to compute type I and type II errors H_0 false H_0 is true Reject H_0 TP=30 (Correct outcome) FP = 30 (Type I error) Fail to R. H_0 FN=10 (Type II error) TN=930 (correct outcome) Many thanks

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